**8.** **In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

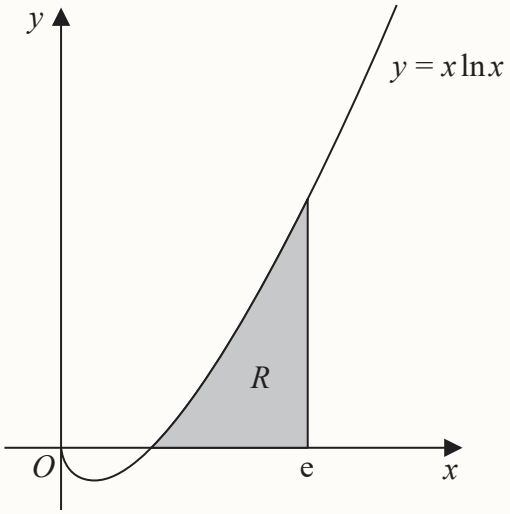



**Figure 3**

(a) Find $\int x^2 \ln x \, dx$

**(3)**

Figure 3 shows a sketch of part of the curve with equation

$$y = x \ln x \qquad x > 0$$

The region $R$, shown shaded in Figure 3, lies entirely above the $x$-axis and is bounded by the curve, the $x$-axis and the line with equation $x = e$.

This region is rotated through $2\pi$ radians about the $x$-axis to form a solid of revolution.

(b) Find the exact volume of the solid formed, giving your answer in simplest form.

**(4)**